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Designing a humanitarian relief logistics network considering the cost of deprivation using a robust-fuzzy-probabilistic planning method

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Abstract

This study addresses deprivation costs in humanitarian logistics by incorporating direct and indirect impacts into a robust-fuzzy-probabilistic framework to optimize resource allocation and minimize delays in disaster response. This model has concepts such as facility location and optimal flow allocation. The results show that with the increase in the uncertainty rate and as a result of the increase in the number of people affected by the accident and the demand for relief items at the scene of the accident, the costs related to the supply and transportation of the items and the injured have increased. Also, by examining the effect of deprivation cost on the results of the model, it was observed that with the increase of deprivation cost, the model aims to increase the goods transferred to satisfy the demand fully. Therefore, the expenses related to deprivation have decreased, but on the other hand, the costs associated with the transfer of products have increased. As a result, the cost of the entire relief logistics network has increased. Finally, by examining the effects of the weighting coefficient of the objective function on the values of the objective function of the problem, it can be seen that with the increase of this weighting coefficient, the value of the objective function has increased due to the increase in the volume of transfers of relief items.

Keywords Logistics of humanitarian aid, Uncertainty, Cost of deprivation, Logistics network

Introduction

Natural disasters occur in different parts of the world in other seasons, leaving many victims and causing much financial damage to governments (Cao et al. 2017). Natural disasters are floods, earthquakes, tsunamis, intense storms, tornadoes, meteorite collisions, etc. Each of these natural events will have numerous harmful consequences

according to their intensity and where they occur. For this reason, preparation for them, planning, forecasting, and necessary preventive measures are vital in these cases (Tatham Christopher 2018). It was said that disasters cause many human and financial losses to societies and nations. Therefore, appropriate emergency logistics planning for natural threats and crisis management is essential. Furthermore, the nature of such incidents is their unexpectedness and unpredictability, despite the progress of science and technology, which makes the availability of preventive plans and emergency response after the occurrence of the incident critical, so the limited resources available to we should allocate the optimal form to the affected people so that their needs are met as much as possible (Salehi et al. 2019).

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Deprivation costs, a critical aspect of humanitarian logistics, encompass direct impacts, such as health deterioration and loss of life, and indirect effects, including long-term societal and economic consequences. These costs are influenced by factors such as response delays, inequitable resource allocation, and the vulnerability of affected populations. By integrating concepts like deprivation cost indices, marginal costs of delay, and probabilistic outcomes under uncertainty, this study aims to capture the nuanced trade-offs in resource distribution and develop a robust, equitable framework for minimizing deprivation in disaster scenarios. Our country is one of the countries where natural disasters such as floods and earthquakes have occurred with various severities and have caused many financial and human losses. Planning for an emergency response to the postdisaster crisis is very important and makes relief operations more efficient and effective. The existence of plans and plans to deal with the consequences of the disaster, along with raising public awareness of the people, will lead to the reduction of human and financial losses, which will be a primary goal of relief responses and reactions. The critical point that should be considered in designing these plans is that the nature of accidents, such as earthquakes, is such that the response should be made at high speed and in a short time. In other words, in such an emergency and complex situation, the decision-maker must quickly and effectively carry out rescue operations and take care of the condition of the injured and injured. To achieve this important goal and take action at the right time, it seems necessary to have access to an efficient and systematic pre-defined program in which all the required activities, their sequence, and connections are specified (Hoseininezhad et al. 2021). Therefore, one of these activities, which is also very important, is the logistics planning for transporting essential items the affected areas need. To achieve the goals of crisis management in accidents, providing aid to the accident areas, the importance of distributing relief goods and picking up the injured is a sensitive and strategic matter because improving the efficiency in carrying and transporting goods and the injured has a significant impact on the rate of rescuers after the accident. Looking at the natural disasters of the last few decades, we clearly understand that the occurrence of natural disasters causes high financial and human costs for governments and societies, and day by day, the concerns in this field are increasing (Jamali et al. 2021). Making the right decisions and taking appropriate and timely actions in each phase of the crisis management cycle leads to the reduction of possible damages at the time of the accident and the reduction of the vulnerability of society. Decision-making tools are needed to make appropriate decisions. In recent years, the losses

and casualties caused by natural and human disasters have increased dramatically, so every year around the world, around 70 thousand people are killed, and nearly 200 million people are affected by these disasters.

Hence, crisis management, especially effective management of relief logistics activities and evacuating the injured from the affected areas, seems necessary. One of the most important activities during a crisis is the logistics of relief items, delivering these items to the injured and transporting the wounded to temporary and permanent medical centers. For this purpose, in this study, a multi-objective logistic model is used for the delivery of priority goods in crisis relief operations so that appropriate decisions can be made for the delivery of relief goods in such a way that the total unfulfilled needs of the goods are minimized. Therefore, the development of last-mile distribution and delivery and evacuation operations are essential strategic issues in relief logistics. Furthermore, it is necessary to provide disaster relief, locate emergency personnel, transfer aid materials after the start of natural disasters, and evacuate injured people from the affected areas to the hospital. All these activities have an extensive effect on the rescue rate of the survivors after the accident. When the damaged places are far away, or it is difficult to access them, or when other means of transportation cannot, helicopters are the most practical means to reach the accident victims, especially flood and earthquake victims.

This research presents a comprehensive model of the humanitarian aid logistics network. In this research, the model presented is to minimize the costs of the entire network, including location, routing, allocation, and maintenance, in which the demand for relief items in the damaged areas and some other essential parameters are considered non-deterministic. Important decisions such as the location of facilities and optimal allocation of flow between facilities to transfer relief items and injured people are made in this model. Considering the existence of uncertainty in the network, the robust-fuzzy-probabilistic programming method has been used to control the non-deterministic parameters of the problem.

Literature review

Every year, natural disasters and unforeseen events such as earthquakes, floods, storms, droughts, and fires plague different parts of the world, and the lack of preparation and proper coping causes heavy losses and damages to nations and their assets, which are sometimes irreparable. Due to the increased number of unexpected events and land, ocean, and atmospheric disasters in recent years, planning to better respond to these events seems necessary. Therefore, one of the most critical actions in responding to a crisis is choosing the right place for

supplies and support from the affected areas. One of the essential measures in the event of accidents is logistical measures and quick relief to crisis areas. Reviewing and evaluating experimental and scientific studies of researchers can open the way for a new method for assessing the concept of relief logistics, which will be discussed in this section.

Mohamadi and Yaghoubi (Mohamadi and Yaghoubi 2017) developed a two-level model for locating transfer points and relief goods distribution centers in earthquake conditions. In the first level, relief facilities are located, and the location of transfer points is determined. In the second level, routing transfers the injured and dead bodies to predetermined points. Also, three scenarios have been considered according to the conditions of the faults in Tehran (Mesha fault, Rey fault, and North fault) with probabilities of 0.35, 0.30, and 0.35. Ultimately, this model was solved using the epsilon constraint method and Gams software due to its multiple goals. The research results indicate 10 points for establishing transfer points along the highways. Rodríguez-Espíndola et al. (Rodríguez-Espíndola et al. 2018) introduced a disaster preparedness system based on multi-objective optimization and geographic information systems for multi-objective decision support. The mapping model is used to prevent the selection of cascading facilities, and a dual-objective optimization model is used to determine the location of emergency facilities, stock forecasting, resource allocation, and aid distribution, as well as the number of actors required to perform these activities. The deprivation cost is often a key economic indicator of human suffering in emergency logistics (Yu et al. 2018). An improved method for adequate and equitable allocation of vital resources in emergency logistics is proposed that considers human suffering using this economic representation.

Torabi et al. (Torabi et al. 2018) designed a scenario-based humanitarian supply chain network model under uncertainty. Their model aimed to reduce the costs of the entire supply chain network, including location, transportation, storage of relief items, and the cost shortage penalty. Their model considered parameters such as fixed construction costs, transportation costs, and demand in a non-deterministic manner and used the scenario-based method. They used GAMS software to solve their model and studied their model in Iran. In a paper, Noham and Tzur (Noham and Tzur 2018) discussed the design of the humanitarian supply chain network using real postdisaster decisions. Reviewing previous articles showed that no comprehensive model can help with post-incident decisions. Therefore, they designed a comprehensive humanitarian supply chain network and 40 numerical examples for their model. They used CPLEX software to solve small- and medium-sized problems and

a forbidden search algorithm to solve larger-size problems. The results obtained from their model showed the high efficiency of the forbidden search algorithm in solving problems of different sizes. Yahya and Bozorgi-Amiri (Yahyaei and Bozorgi-Amiri 2018) designed a supply chain network under uncertainty. The main goal of this article was to control the failure of distribution centers to meet the needs of demand centers by reducing investment costs. Also, they used the robust optimization method to control the uncertain parameters and showed that the total investment costs increase with the increase of the uncertainty level. Elçi and Noyan (Elçi and Noyan 2018) modeled a scenario-based humanitarian supply chain network. Their model investigated the optimal number and location of facilities, facility capacity, inventory levels, and transportation network conditions under demand uncertainty after the accident. They used a branch-and-cut algorithm based on the Benders decomposition to solve their probabilistic model. They demonstrated the effectiveness of their model by studying hurricane threats in the southeastern USA. Paul and Wang (Paul and Wang 2019) addressed robust modeling of the location-assignment problem in conditions. This research aims to minimize the total social costs, including initial setup and procurement and deprivation costs associated with delays in accessing resources. The models include various uncertainties in earthquake magnitude, such as facility damage, losses by intensity, and travel time. Zanganeh et al. (Zanganeh et al. 2019) presented a dual-objective model for humanitarian logistics networks in post-crisis situations. The purpose of this model was to maximize the functional performance of saving the lives of injured people and to help injured people with a delay. Hong (Hong 2020) addressed humanitarian logistics network design with facility location decisions. He used an ideal planning approach to solve his four-objective model. Data coverage research methods were used to evaluate effective answers. Mansoori et al. (Mansoori et al. 2020) proposed a multi-objective mathematical model for the humanitarian supply chain network design problem that simultaneously minimizes (1) the number of injured not transported to hospitals and the total number of homeless not evacuated from the affected area and (2) pays for the total unmet needs of auxiliary goods. This mathematical model used robust optimization to control demand parameters and travel time. Ghaffari et al. (Ghaffari et al. 2020) modeled an emergency logistics network to optimize the costs of the entire network. This problem also considers the scheduling of multi-stage operations with limited capacity. The exact method and particle swarm optimization algorithm have been used to solve the problem. Cheng et al. (Cheng et al. 2021) examined the issue of humanitarian

aid food distribution. In this model, a group of food donors have taken steps to meet the needs of the affected areas, and the goal is fair distribution among them. They used the robust optimization method to control the non-deterministic demand parameter. Hoseininezhad et al. (Hoseininezhad et al. 2021) presented a multi-objective mathematical model for pre-positioning a rescue chain for road accidents. They used the NSGA II algorithm to solve their multi-objective model. In this model, using the obtained results, they suggested potential locations for the relief chain. Akbari et al. (Akbari et al. 2021) modeled and solved a relief logistics problem by establishing coordination relationships in the supply chain network. Their main goal in this research was to minimize the total costs of network design. They used a genetic algorithm to solve the model. Boostani et al. (Boostani et al. 2021) proposed a three-level aid supply chain network before and after an incident to improve sustainability in humanitarian aid operations. Their goal was to minimize the costs of the entire humanitarian supply chain, maximize social welfare by maximizing the minimum level of satisfaction in the affected areas, and minimize the amount of greenhouse gas emissions. They used CP and LOM methods to solve their three-objective model.

Shu et al. (Shu et al. 2022) designed a humanitarian relief network with uncertain demand for each affected area. The mathematical model aimed to maximize the response to the demand of the damaged points. They conducted a case study on Typhoon Rammasun in 2014. Computational results demonstrate the effectiveness of the solution approaches and show that the stochastic programming models with chance constraints are superior to the deterministic model for humanitarian relief network design. Modiri et al. (Modiri et al. 2022) presented a multi-objective mixed integer mathematical programming model for the design of relief item distribution networks in sustainable disaster relief logistics. The first objective function minimizes the total network costs. These are divided into two parts: (1) relief costs, including transportation costs, inventory costs, and fixed facility costs, and (2) social costs (deprivation cost). The second objective function minimizes the amount of pollution produced by the network. To show the validity of the proposed model, a case study based on real data (the 2019 flood in Sari City, Mazandaran province) was evaluated. Using the proposed model, decision-makers and managers can make strategic and tactical decisions with the least cost and time, and in relief planning, they can improve the structure of distribution and inventory networks and reduce the dissatisfaction of victims. Ghasemi et al. (Ghasemi et al. 2022) presented a multi-objective

humanitarian aid logistics model under probabilistic planning optimization. The objectives of this paper are (1) to minimize the expected value of the total costs of the relief supply chain, (2) to minimize the maximum number of unmet demands for relief workers, and (3) to minimize the total probability of failed evacuations along the routes. This paper proposes a scenario-based stochastic multi-objective location-allocation-routing model for a real humanitarian relief logistics problem, focusing on pre- and postdisaster situations in uncertainty. A simulation approach was used to deal with demand uncertainty. The presented model has been solved using the epsilon-constraint method for minor and medium-scale problems and three metaheuristic algorithms for large-scale problems (case study). Experimental results show that this model can be used to locate shelters and aid distribution centers, determine appropriate routes, and allocate resources in uncertain and real disaster situations. Abazari et al. (Abazari et al. 2022) modeled a humanitarian supply chain network that includes locating relief centers, pre-positioning and distributing relief items, and providing medical treatment in affected areas to help affected people. The goal of this model is to minimize total costs. The problem is formulated as a complex integer linear mathematical model. A two-stage stochastic fuzzy hybrid approach is developed to capture uncertainty. The grasshopper optimization algorithm has been used to solve the model on a large scale, and its performance has been investigated through computational experiments.

Deprivation costs play a critical role in humanitarian logistics, measuring the human suffering caused by unmet needs in disaster scenarios. These costs, encompassing direct impacts like mortality and indirect effects such as long-term economic losses, are key to designing effective disaster response frameworks (Holguín-Veras et al. 2013). Recent studies highlight the importance of integrating deprivation costs into humanitarian logistics models. For instance, Silva and Leiras (Silva and Leiras 2021) conducted a comprehensive review underscoring their significance in optimizing resource allocation and minimizing suffering in disaster zones. Similarly, van Steenbergen et al. (Steenbergen et al. 2023) proposed a stochastic dynamic model incorporating deprivation costs, demonstrating its efficacy in improving disaster response operations. Advanced modeling techniques have been utilized to address uncertainties inherent in disaster environments. Boonmee et al. (Boonmee et al. 2020) introduced a multi-objective fuzzy mathematical programming model, effectively managing the ambiguity in demand and supply data. Further, Sharma and Mehta

(Sharma and Mehta 2021) developed a probabilistic fuzzy goal programming model for optimizing postdisaster supply chain, accommodating demand and transportation cost uncertainties. Integrating robust, fuzzy, and probabilistic approaches enhances the resilience of relief operations, ensuring the timely and equitable allocation of resources and reducing deprivation costs.

Based on the studied literature, research innovation can be expressed in the following cases:

- Development of a relief logistics network model in pre- and post-crisis conditions
- Using robust-fuzzy-probabilistic programming method to control non-deterministic parameters

Problem definition and modeling

This section discusses the modeling of the humanitarian aid logistics problem. According to Fig. 1, in this problem, a set of suppliers have the task of providing environmentally friendly and non-environmentally friendly relief items. There is also a set of potential locations for aid distribution centers, which should be located according to the severity of the crisis and be considered in different scenarios. In this case, the distribution centers, as an intermediary, transfer relief items from the supplier to the demand points after the crisis. In the meantime, due to the size of the crisis, there is a possibility of injury to some people, who should be transferred to temporary emergency centers in case of minor injuries and to hospitals in case of serious injuries. There is also the possibility of a severe crisis after transferring people to emergency centers, which will also be addressed in the upcoming modeling. On the other hand, some people need to go to shelters after an accident.

Due to the uncertainty of the amount of demand at the place of the accident and the severity of the crisis, the parameters of demand and cost of transmission have been considered uncertain in different scenarios. Therefore, if any of the distribution centers are out of reach due to the disruption, it is impossible to transfer relief items from that center, and other centers should be used to meet the demand of the disaster areas.

In the designed model, there are two different types of decisions. First, strategic decisions are essential, such as choosing the optimal number and location of relief item distribution centers and potential emergency centers. Also, tactical decisions, such as how to transfer, buy, and store environmentally compatible and non-compatible relief items and how to transport the injured by vehicle, are of great importance.

Considering the above, the assumptions of the problem can be stated as follows:

- The severity of the crisis is considered based on different scenarios, each with a different probability of occurrence.
- The location of suppliers and hospitals is already known.
- Demand and transmission costs are considered non-deterministic and trapezoidal fuzzy numbers.
- Aid items are considered to be compatible with the environment and non-compatible with it.
- The capacity of the facility and the vehicles is known in advance.

According to the above assumptions and stated objective functions, the symbols used in the modeling are as follows.

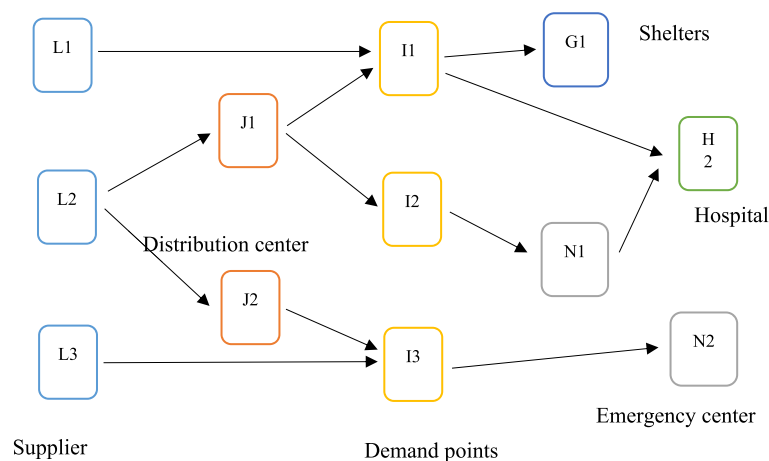


Fig. 1 Proposed humanitarian aid logistics network

Sets		Sets	
I	Set of demand points $i \in I$	Tr_{ljcs}	The cost of transporting relief goods $c \in C$ from the supplier $l \in L$ to the relief distribution center $j \in J$ after the disaster in the scenario $s \in S$
J	The set of potential distribution centers $j \in J$	Tr_{lics}	The cost of transporting relief goods $c \in C$ from the supplier $l \in L$ to the demand point $i \in I$ after the disaster in the scenario $s \in S$
K	Capacity set of aid distribution centers $k \in K$	Tr_{jics}	The cost of transporting relief goods $c \in C$ from the relief distribution center $j \in J$ to the demand point $i \in I$ after the disaster in scenario $s \in S$
L	The set of suppliers $l \in L$	Tr_{ihc}	The cost of transporting the injured from the demand point $i \in I$ to the hospital $h \in H$ after the disaster in the scenario $s \in S$
C	Relief goods set $c \in C$	Tr_{ins}	The cost of transporting the injured from the demand point $i \in I$ to the temporary emergency center $n \in N$ after the disaster in the scenario $s \in S$
C_1	A set of items with environmentally friendly packaging $c_1 \in C_1$	Tr_{nhs}	The cost of transporting the injured from the temporary emergency center $n \in N$ to the hospital $h \in H$ after the disaster in the scenario $s \in S$
C_2	A set of items with non-environmentally friendly packaging $c_2 \in C_2$	Tr_{igs}	The cost of transporting the injured from the demand point $i \in I$ to the shelter $g \in G$ after the disaster in the scenario $s \in S$
S	The set of probability events of the scenario $s \in S$	h_{jc}	The cost of maintaining a unit of relief goods $c \in C$ in the relief distribution center $j \in J$
N	The set of potential emergency centers $n \in N$	v_{lc_1}	The amount of relief goods $c_1 \in C_1$ that can be provided from supplier $l \in L$
H	Hospital set $h \in H$	v_{lc_2}	The amount of relief goods $c_2 \in C_2$ that can be provided from supplier $l \in L$
G	The set of shelters $g \in G$	v_{lc_1s}	The amount of relief goods $c_1 \in C_1$ that can be provided from supplier $l \in L$ in scenario $s \in S$
Parameters		v_{lc_2s}	The amount of relief goods $c_2 \in C_2$ that can be provided from supplier $l \in L$ in scenario $s \in S$
p_s	The probability of scenario $s \in S$	V_k	The amount of capacity for each aid distribution center $k \in K$
d_{ics}	Demand amount of relief items $c \in C$ at demand point $i \in I$ in scenario $s \in S$	CC_{ic}	The cost of denying demand point $i \in I$ from receiving relief goods $c \in C$
w_{is}	The number of injured persons required to be treated at the demand point $i \in I$ in the scenario $s \in S$	δ_n	The percentage of injured people who should be transferred to the hospital from potential emergency center $n \in N$
θ_c	Space required to store relief goods $c \in C$ in cubic meters		
F_{jk}	The fixed cost of establishing a relief distribution center in location $j \in J$ with capacity level $k \in K$		
F_n	The fixed cost of establishing a temporary emergency center in place $n \in N$		
b_{lc_1}	The cost of buying a unit of relief goods with environmentally friendly packaging $c_1 \in C_1$ from supplier $l \in L$		
b_{lc_2}	The cost of buying a unit of relief goods with non-environmentally friendly packaging $c_2 \in C_2$ from supplier $l \in L$		
b_{lc_1s}	The cost of purchasing a unit of relief goods with environmentally friendly packaging $c_1 \in C_1$ from supplier $l \in L$ in scenario $s \in S$		
b_{lc_2s}	The cost of purchasing a unit of relief goods with non-environmentally friendly packaging $c_2 \in C_2$ from supplier $l \in L$ in scenario $s \in S$		
Tr_{ljc}	The cost of transporting relief goods $c \in C$ from the supplier $l \in L$ to the relief distribution center $j \in J$		

Sets	
ω_i	Percentage of people in need of shelter from demand point $i \in I$
ωP_i	The percentage of people who need an emergency center from the demand point $i \in I$
$CapH_h$	Hospital capacity $h \in H$
$CapG_g$	Shelter capacity $g \in G$
$CapN_n$	Potential temporary emergency center capacity $n \in N$
Decision variables	
I_{jc}	The amount of relief goods $c \in C$ stored in the relief distribution center $j \in J$ in the stage before the crisis
I_{jcs}	The amount of relief goods $c \in C$ stored in the relief distribution center $j \in J$ in the post-crisis phase in the scenario $s \in S$
X_{jk}	It takes the value of 1 if the relief distribution center with capacity $k \in K$ is opened at the potential location $j \in J$, and it takes the value of 0 otherwise
Y_{jics}	The amount of relief goods $c \in C$ transferred from the aid distribution center $j \in J$ to the demand point $i \in I$ in the scenario $s \in S$
Y_{lics}	The amount of relief goods $c \in C$ transferred from the supplier $l \in L$ to the demand point $i \in I$ in the scenario $s \in S$
Y_{lic1s}	The amount of relief goods $c_1 \in C_1$ transferred from supplier $l \in L$ to demand point $i \in I$ in scenario $s \in S$
Y_{lic2s}	The amount of relief goods $c_2 \in C_2$ transferred from the supplier $l \in L$ to the demand point $i \in I$ in the scenario $s \in S$
Q_{ljc1}	Order amount of a relief item with environmentally friendly packaging $c_1 \in C_1$ from supplier $l \in L$ by relief distribution center $j \in J$
Q_{ljc2}	Order amount of a relief item with non-environmentally friendly packaging $c_2 \in C_2$ from supplier $l \in L$ by relief distribution center $j \in J$
Q_{ljc1s}	Order amount of a relief item with environmentally friendly packaging $c_1 \in C_1$ from supplier $l \in L$ by relief distribution center $j \in J$ in scenario $s \in S$
Q_{ljc2s}	Order amount of a relief item with non-environmentally friendly packaging $c_2 \in C_2$ from supplier $l \in L$ by relief distribution center $j \in J$ in scenario $s \in S$
Q_{ljc}	Amount of relief items $c \in C$ purchased from supplier $l \in L$ by relief distribution center $j \in J$ in the pre-disaster phase

Sets	
Q_{ljcs}	Amount of relief items $c \in C$ purchased from supplier $l \in L$ by relief distribution center $j \in J$ in the post-disaster stage in scenario $s \in S$
XN_{ins}	The number of people with minor injuries transferred from the demand point $i \in I$ to the temporary emergency center $n \in N$ in the scenario $s \in S$
XH_{ihhs}	The number of people with serious injuries transferred from demand point $i \in I$ to hospital $h \in H$ in scenario $s \in S$
NH_{nhhs}	The number of seriously injured people transferred from temporary emergency center $n \in N$ to hospital $h \in H$ in scenario $s \in S$
XG_{igs}	Number of people transferred from demand point $i \in I$ to shelter $g \in G$ in scenario $s \in S$
U_n	It takes the value of 1 if the temporary emergency center is opened at the location $n \in N$ and otherwise it takes the value of 0

According to the symbols defined in the previous section, the mathematical model of humanitarian aid logistics network under uncertainty conditions will be as follows.

$$\begin{aligned}
 \text{Min} Z_1 = & \sum_{j=1}^J \sum_{k=1}^K F_{jk} \cdot X_{jk} + \sum_{n=1}^N F_n \cdot U_n + \sum_{j=1}^J \sum_{c=1}^C h_{jc} \cdot I_{jc} + \sum_{j=1}^J \sum_{l=1}^L \sum_{c_1=1}^{C_1} b_{lc_1} \cdot Q_{ljc_1} \\
 & + \sum_{j=1}^J \sum_{l=1}^L \sum_{c_2=1}^{C_2} b_{lc_2} \cdot Q_{ljc_2} + \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C Tr_{ljc} \cdot Q_{ljc} \\
 & + \sum_{s=1}^S P_s \left[\sum_{j=1}^J \sum_{c=1}^C h_{jc} \left(I_{jcs} - \sum_{i=1}^I Y_{jics} \right) + \sum_{j=1}^J \sum_{l=1}^L \sum_{c_1=1}^{C_1} b_{lc_1s} \cdot Q_{ljc_1s} \right. \\
 & + \sum_{j=1}^J \sum_{l=1}^L \sum_{c_2=1}^{C_2} b_{lc_2s} \cdot Q_{ljc_2s} + \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C Tr_{ljcs} \cdot Q_{ljcs} \\
 & + \sum_{i=1}^I \sum_{l=1}^L \sum_{c=1}^C Tr_{lics} \cdot Y_{lics} + \sum_{i=1}^I \sum_{j=1}^J \sum_{c=1}^C Tr_{jics} \cdot Y_{jics} \\
 & + \sum_{i=1}^I \sum_{h=1}^H Tr_{ihis} \cdot XH_{ihhs} + \sum_{i=1}^I \sum_{n=1}^N Tr_{ins} \cdot XN_{ins} \\
 & + \sum_{n=1}^N \sum_{h=1}^H Tr_{nhhs} \cdot NH_{nhhs} + \sum_{i=1}^I \sum_{g=1}^G Tr_{igs} \cdot XG_{igs} \\
 & \left. + \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C Cc_{ic} \cdot (d_{ics} - Y_{jics}) + \sum_{l=1}^L \sum_{i=1}^I \sum_{c=1}^C Cc_{ic} \cdot (d_{ics} - Y_{lics}) \right] \quad (1)
 \end{aligned}$$

s.t. :

$$Q_{ljc} = Q_{ljc1} + Q_{ljc2}, \forall j, l, c, c_1, c_2 \quad (2)$$

$$Q_{ljcs} = Q_{ljc1s} + Q_{ljc2s}, \forall j, l, c, c_1, c_2, s \quad (3)$$

$$I_{jc} = \sum_{l=1}^L Q_{ljc}, \forall j, c \quad (4)$$

$$I_{jcs} = \sum_{l=1}^L Q_{ljcs} + \sum_{l=1}^L Q_{ljc_1} + \sum_{l=1}^L Q_{ljc_2}, \forall j, c, c_1, c_2, s \quad (5)$$

$$\sum_{i=1}^I Y_{jics} \leq I_{jcs}, \forall j, c, s \quad (6)$$

$$\sum_{c=1}^C \theta_c \cdot I_{jc} \leq \sum_{k=1}^K V_k \cdot X_{jk}, \forall j \quad (7)$$

$$\sum_{c=1}^C \theta_c \cdot I_{jcs} \leq \sum_{k=1}^K V_k \cdot X_{jk}, \forall j, s \quad (8)$$

$$\sum_{k=1}^K X_{jk} \leq 1, \forall j \quad (9)$$

$$\sum_{j=1}^J Q_{ljc_1} \leq v_{lc_1}, \forall l, c_1 \quad (10)$$

$$\sum_{j=1}^J Q_{ljc_2} \leq v_{lc_2}, \forall l, c_2 \quad (11)$$

$$\sum_{j=1}^J Q_{ljc_1s} + \sum_{i=1}^I Y_{lic_1s} \leq v_{lc_1s}, \forall l, c_1, s \quad (12)$$

$$\sum_{j=1}^J Q_{ljc_2s} + \sum_{i=1}^I Y_{lic_2s} \leq v_{lc_2s}, \forall l, c_2, s \quad (13)$$

$$Y_{lics} = Y_{lic_1s} + Y_{lic_2s}, \forall i, l, c, c_1, c_2, s \quad (14)$$

$$\sum_{n=1}^N XN_{ins} + \sum_{h=1}^H XH_{ihis} + \sum_{g=1}^G XG_{igs} = w_{is}, \forall i, s \quad (15)$$

$$\sum_{g=1}^G XG_{igs} = \omega_i \cdot w_{is}, \forall i, s \quad (16)$$

$$\sum_{n=1}^N XN_{ins} = \omega p_i \cdot w_{is}, \forall i, s \quad (17)$$

$$\delta_n \sum_{i=1}^I XN_{ins} = \sum_{h=1}^H NH_{nhis}, \forall n, s \quad (18)$$

$$\sum_{h=1}^H NH_{nhis} \leq BM \cdot U_n, \forall n, s \quad (19)$$

$$\sum_{i=1}^I XG_{igs} \leq CapG_g, \forall g, s \quad (20)$$

$$\sum_{n=1}^N NH_{nhis} + \sum_{i=1}^I XH_{ihis} \leq CapH_h, \forall h, s \quad (21)$$

$$\sum_{i=1}^I XN_{ins} \leq CapN_n \cdot U_n, \forall n, s \quad (22)$$

$$\begin{aligned} &I_{jc}, I_{jcs}, Y_{jics}, Y_{lics}, Y_{lic_1s}, Y_{lic_2s}, Q_{ljc_1}, Q_{ljc_2}, \\ &Q_{ljc_1s}, Q_{ljc_2s}, Q_{ljcs}, Q_{ljcs}, XN_{ins}, XH_{ihis}, NH_{nhis}, XG_{igs} \geq 0, \text{integer} \\ &U_n, X_{jk} \in \{0,1\} \end{aligned} \quad (23)$$

Equation (1) shows the costs of the entire humanitarian logistics network. These costs include the total cost of location, purchase, transfer, allocation, and exclusion costs. Equations (2) and (3), respectively, show the total amount of goods purchased from the supplier by the relief distribution center in the conditions before and after the crisis. Equation (4) shows the inventory level of aid items purchased before the crisis, and Eq. (5) shows the inventory level that is compatible/incompatible with the environment after the crisis. Equation (6) shows that the number of relief items transferred from the distribution centers to the demand points should be less than the amount of stock in the distribution center's warehouse. Equations (7) and (8) show the capacity limitation of relief items distribution centers in all capacity levels. Equation (9) guarantees that each distribution center can use at most one capacity level. Relationships (10) to (13) show the constraints related to the capacity to provide environmentally friendly/unfriendly relief items. Equation (14) shows the total relief items transferred from the supplier to the demand points. The relation (15) guarantees that all injured people should

be transferred to temporary hospitals or emergency rooms. The relation (16) shows the percentage of people with the demand points that should be transferred to the shelter. Equation (17) shows the number of people needing the emergency center from the demand point. The relation (18) shows the percentage of the injured who must be transferred from the emergency center to the hospital for treatment. Relationship (19) shows that if an emergency center is established, that center can be used to transfer the injured to hospitals. Equation (20) shows the capacity of using shelters. Relationships (21) and (22) show the capacity of emergency centers and hospitals. The relation (23) shows the type and gender of the decision-making variables.

Due to the dynamic and fluctuating nature of some important parameters (including transportation costs and demand) that are beyond planning, as well as the unavailability and unattainability of historical data required in the design phase, these parameters are mainly based on experts' opinions and subjective experiences. It is estimated. Therefore, the above fuzzy parameters are formulated as non-deterministic data in the form of trapezoidal fuzzy numbers. It is worth noting that evaluating the transportation cost and exact demand for long-term decisions is complicated and sometimes even impossible. Even if one can estimate a probability distribution function for these parameters, it is possible that these parameters do not behave the same as the past data. Therefore, these parameters that change in a long-term planning horizon are considered fuzzy data.

In addition, for the non-deterministic (probable) constraints with non-deterministic data on the left or right side of their equality, the non-deterministic constraint programming method is usually used. Suppose this method controls the confidence level of establishing these uncertain limits. In that case, the concept of decision-making can obtain the minimum confidence level as a suitable safe margin for establishing each of these limits. To do this, two measures of the standard fuzzy method, optimistic and pessimistic fuzzy, are commonly used. It is worth mentioning that the optimistic fuzzy represents the level of optimistic probability of the occurrence of an uncertain event, including uncertain parameters. In contrast, the pessimistic fuzzy represents a pessimistic decision about an uncertain event. However, it is more conservative to use a pessimistic hybrid fuzzy; we assume that decision-making has an optimistic attitude to establish non-deterministic limits. Now, an axiomatic equivalent of the original non-deterministic model can be formulated based on the mentioned fuzzy parameters and using the expected value for the objective

function and the pessimistic action for the non-deterministic constraints. To do this, first consider the abbreviated form of the presented model:

$$\begin{aligned} \min Z &= F.y + \tilde{C}_s.x_s \\ \text{s.t. :} \\ Ax_s &\geq \tilde{D}_s \\ Ex_s &\leq Sy \\ y &\in \{0,1\}, x_s \geq 0 \end{aligned} \quad (24)$$

where the vectors $\tilde{D}_s, \tilde{C}_s, F$, and S represent the fixed construction cost, variable cost (transportation), demand amount, and facility capacity. Also, A and E are the matrix of coefficients, and finally x_s and y are continuous variables and zero and one under scenario s , respectively. Now, it is assumed that the vectors $\tilde{D}_s$ and $\tilde{C}_s$ in the above model are presented as non-deterministic parameters. According to the general form of non-deterministic limited programming, the expected value of the objective function and the pessimistic fuzzy should be obtained to deal with the objective function and the non-deterministic constraint, respectively. Now, according to the abbreviated form, the pessimistic fuzzy basic model is as follows:

$$\begin{aligned} \min Z &= E[Z] = F.y + E[\tilde{C}_s].x_s \\ \text{s.t. :} \\ NEC \left\{ Ax_s \geq \tilde{D}_s \right\} &\geq \alpha_s \\ Ex_s &\leq Sy \\ y &\in \{0,1\}, x_s \geq 0 \end{aligned} \quad (25)$$

where α_s controls the minimum degree of certainty of establishing the non-deterministic limit with a (pessimistic) decision-making approach. According to the trapezoidal probability distribution for fuzzy parameters, the general form of relations (25) is as follows:

$$\begin{aligned} \min Z &= F.y + \left(\frac{C_s^1 + C_s^2 + C_s^3 + C_s^4}{4} \right) x_s \\ \text{s.t. :} \\ Ax_s &\geq (1 - \alpha_s) D_s^3 + \alpha_s D_s^4 \\ Ex_s &\leq Sy \\ y &\in \{0,1\}, x_s \geq 0 \end{aligned} \quad (26)$$

In probabilistic models, the minimum confidence level to establish an uncertain constraint should be determined by decision preferences. However, as seen in the presented models, the objective function is not sensitive to the deviation from its expected value, which means that achieving stable solutions in the basic model is not guaranteed. In such cases, a high risk may be imposed on decision-making in many real cases, especially in

strategic decisions where the stabilization of the solution is critical to a large extent. Therefore, to deal with this inefficient situation, the non-deterministic robust-probabilistic programming approach is used. This approach takes advantage of the advantages of both robust and probabilistic planning, which clearly makes it different from other uncertainty planning approaches. In this research, uncertainty planning is applied based on the presented model, which is as follows:

$$\begin{aligned} \min \omega_1 &= E[Z] + \xi' (Z_{(\max)} - Z_{(\min)}) + \omega \sum_s P_s \{E[Z] - E[Z_s] + 2\theta_s\} \\ &\quad + \eta \sum_s P_s \left[D_s^4 - \frac{(\alpha_s - \lambda) D_s^4 + (1 - \alpha_s) D_s^3}{1 - \lambda} \right] + \\ &\quad \text{s.t. :} \\ &\quad A x_s \geq (1 - \alpha_s) D_s^3 + \alpha_s D_s^4 \\ &\quad E x_s \leq S y \\ &\quad y \in \{0,1\}, x_s \geq 0, \alpha_s \in [0.1, 0.9] \end{aligned} \quad (27)$$

where M is a vast non-negative number and $Z_{(\max)}$ and $Z_{(\min)}$ and $E[Z_s]$ can be expressed as follows:

$$\begin{aligned} Z_{(\max)} &= F \cdot y + \sum_s P_s C_s^4 x_s \\ Z_{(\min)} &= F \cdot y + \sum_s P_s C_s^1 x_s \\ E[Z_s] &= F \cdot y + \left[\frac{1-\lambda}{2} (C_s^1 + C_s^2) + \frac{\lambda}{2} (C_s^3 + C_s^4) \right] x_s \end{aligned} \quad (28)$$

In the first objective function of Eq. (27), the first term refers to the expected value of the first objective function using the average values of the uncertain parameters of the model. The second term refers to the penalty cost for deviation from the expected value of the first objective function (stability of optimality). The third sentence also shows the total cost of the penalty for deviation from the demand (non-deterministic parameter). Therefore, the parameter ξ' of the weighting coefficient of the objective function, η is the penalty cost of not estimating the demand. α_s parameters as correction coefficients in the value of fuzzy levels show the numbers, which should be a number between 0.1 and 0.9.

$$\begin{aligned} \min Z_1 &= \xi' (Z_{\max} - Z_{\min}) + \omega \sum_s P_s \{E[Z] - E[Z_s] + 2\theta_s\} \\ &\quad + \eta \sum_{i=1}^I \sum_{c=1}^C \sum_{s=1}^S \left[d_{ics}^4 - \frac{(\alpha_s - \lambda) d_{ics}^4 + (1 - \alpha_s) d_{ics}^3}{1 - \lambda} \right] \end{aligned} \quad (29)$$

s.t. :

$$Eqs(2) - (23) \quad (30)$$

$$\begin{aligned} Z_{\min} &= \sum_{j=1}^J \sum_{k=1}^K F_{jk} \cdot X_{jk} + \sum_{n=1}^N F_n \cdot U_n + \sum_{j=1}^J \sum_{c=1}^C h_{jc} \cdot I_{jc} + \sum_{j=1}^J \sum_{l=1}^L \sum_{c_1=1}^{C_1} b_{lc_1} \cdot Q_{ljc_1} + \\ &\quad \sum_{j=1}^J \sum_{l=1}^L \sum_{c_2=1}^{C_2} b_{lc_2} \cdot Q_{ljc_2} + \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C \sum_{v_1=1}^{V_1} Tr_{ljcv_1} \cdot Q_{ljc} + \\ &\quad \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{c=1}^C h_{jc} \left(I_{jcs} - \sum_{i=1}^I Y_{jics} \right) + \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c_1=1}^{C_1} b_{lc_1} \cdot Q_{ljc_1} + \\ &\quad \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c_2=1}^{C_2} b_{lc_2} \cdot Q_{ljc_2} + \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C \sum_{v_1=1}^{V_1} Tr_{ljcv_1} \cdot Q_{ljcs} + \\ &\quad \sum_{s=1}^S P_s \sum_{i=1}^I \sum_{l=1}^L \sum_{c=1}^C Tr_{ilcs}^1 \cdot Y_{ilcs} + \sum_{s=1}^S P_s \sum_{i=1}^I \sum_{j=1}^J \sum_{c=1}^C Tr_{ijcs}^1 \cdot Y_{ijcs} + \\ &\quad \sum_{s=1}^S P_s \sum_{i=1}^I \sum_{h=1}^H Tr_{ihls}^1 \cdot XH_{ihls} + \sum_{s=1}^S P_s \sum_{i=1}^I \sum_{n=1}^N Tr_{ihns}^1 \cdot XN_{ihns} + \\ &\quad \sum_{s=1}^S P_s \sum_{n=1}^N \sum_{h=1}^H Tr_{nhls}^1 \cdot NH_{nhls} + \sum_{s=1}^S P_s \sum_{i=1}^I \sum_{g=1}^G Tr_{igls}^1 \cdot XG_{igls} + \\ &\quad \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{i=1}^I \sum_{c=1}^C Cc_{ic} \cdot \left(\left[(1 - \alpha_s) d_{ics}^3 + \alpha_s d_{ics}^4 \right] - Y_{jics} \right) + \\ &\quad \sum_{s=1}^S P_s \sum_{l=1}^L \sum_{i=1}^I \sum_{c=1}^C Cc_{ic} \cdot \left(\left[(1 - \alpha_s) d_{ics}^3 + \alpha_s d_{ics}^4 \right] - Y_{ilcs} \right) \end{aligned} \quad (31)$$

$$\begin{aligned} Z_{\max} &= \sum_{j=1}^J \sum_{k=1}^K F_{jk} \cdot X_{jk} + \sum_{n=1}^N F_n \cdot U_n + \sum_{j=1}^J \sum_{c=1}^C h_{jc} \cdot I_{jc} + \sum_{j=1}^J \sum_{l=1}^L \sum_{c_1=1}^{C_1} b_{lc_1} \cdot Q_{ljc_1} + \\ &\quad \sum_{j=1}^J \sum_{l=1}^L \sum_{c_2=1}^{C_2} b_{lc_2} \cdot Q_{ljc_2} + \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C \sum_{v_1=1}^{V_1} Tr_{ljcv_1} \cdot Q_{ljc} + \\ &\quad \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{c=1}^C h_{jc} \left(I_{jcs} - \sum_{i=1}^I Y_{jics} \right) + \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c_1=1}^{C_1} b_{lc_1} \cdot Q_{ljc_1} + \\ &\quad \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c_2=1}^{C_2} b_{lc_2} \cdot Q_{ljc_2} + \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C \sum_{v_1=1}^{V_1} Tr_{ljcv_1} \cdot Q_{ljcs} + \\ &\quad \sum_{s=1}^S P_s \sum_{i=1}^I \sum_{l=1}^L \sum_{c=1}^C Tr_{ilcs}^4 \cdot Y_{ilcs} + \sum_{s=1}^S P_s \sum_{i=1}^I \sum_{j=1}^J \sum_{c=1}^C Tr_{ijcs}^4 \cdot Y_{ijcs} + \\ &\quad \sum_{s=1}^S P_s \sum_{i=1}^I \sum_{h=1}^H Tr_{ihls}^4 \cdot XH_{ihls} + \sum_{s=1}^S P_s \sum_{i=1}^I \sum_{n=1}^N Tr_{ihns}^4 \cdot XN_{ihns} + \\ &\quad \sum_{s=1}^S P_s \sum_{n=1}^N \sum_{h=1}^H Tr_{nhls}^4 \cdot NH_{nhls} + \sum_{s=1}^S P_s \sum_{i=1}^I \sum_{g=1}^G Tr_{igls}^4 \cdot XG_{igls} + \\ &\quad \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{i=1}^I \sum_{c=1}^C Cc_{ic} \cdot \left(\left[(1 - \alpha_s) d_{ics}^3 + \alpha_s d_{ics}^4 \right] - Y_{jics} \right) + \\ &\quad \sum_{s=1}^S P_s \sum_{l=1}^L \sum_{i=1}^I \sum_{c=1}^C Cc_{ic} \cdot \left(\left[(1 - \alpha_s) d_{ics}^3 + \alpha_s d_{ics}^4 \right] - Y_{ilcs} \right) \end{aligned} \quad (32)$$

$$\begin{aligned}
E[Z] = & \sum_{j=1}^J \sum_{k=1}^K F_{jk} \cdot X_{jk} + \sum_{n=1}^N F_n \cdot U_n + \sum_{j=1}^J \sum_{c=1}^C h_{jc} \cdot J_{jc} + \sum_{j=1}^J \sum_{l=1}^L \sum_{c_1=1}^{C_1} b_{lc_1} \cdot Q_{ljc_1} + \\
& \sum_{j=1}^J \sum_{l=1}^L \sum_{c_2=1}^{C_2} b_{lc_2} \cdot Q_{ljc_2} + \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C Tr_{ljc} \cdot Q_{ljc} + \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c_2=1}^{C_2} b_{lc_2s} \cdot Q_{ljc_2s} + \\
& \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{c=1}^C h_{jc} \left(I_{jcs} - \sum_{i=1}^I Y_{jics} \right) + \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c_1=1}^{C_1} b_{lc_1s} \cdot Q_{ljc_1s} + \\
& \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C \left[\frac{Tr_{ljcs}^1 + Tr_{ljcs}^2 + Tr_{ljcs}^3 + Tr_{ljcs}^4}{4} \right] \cdot Q_{ljcs} + \\
& \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C \left[\frac{Tr_{ljcs}^1 + Tr_{ljcs}^2 + Tr_{ljcs}^3 + Tr_{ljcs}^4}{4} \right] \cdot Y_{ljcs} + \\
& \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C \left[\frac{Tr_{ljcs}^1 + Tr_{ljcs}^2 + Tr_{ljcs}^3 + Tr_{ljcs}^4}{4} \right] \cdot Y_{jics} + \\
& \sum_{s=1}^S P_s \sum_{i=1}^I \sum_{h=1}^H \left[\frac{Tr_{ihs}^1 + Tr_{ihs}^2 + Tr_{ihs}^3 + Tr_{ihs}^4}{4} \right] \cdot XH_{ihs} \\
& + \sum_{s=1}^S P_s \sum_{i=1}^I \sum_{n=1}^N \left[\frac{Tr_{ins}^1 + Tr_{ins}^2 + Tr_{ins}^3 + Tr_{ins}^4}{4} \right] \cdot XN_{ins} + \\
& \sum_{s=1}^S P_s \sum_{n=1}^N \sum_{h=1}^H \left[\frac{Tr_{nhs}^1 + Tr_{nhs}^2 + Tr_{nhs}^3 + Tr_{nhs}^4}{4} \right] \cdot NH_{nhs} + \\
& \sum_{s=1}^S P_s \sum_{i=1}^I \sum_{g=1}^G \left[\frac{Tr_{igs}^1 + Tr_{igs}^2 + Tr_{igs}^3 + Tr_{igs}^4}{4} \right] \cdot XG_{igs} + \\
& \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C Cc_{lc} \cdot \left((1 - \alpha_s) d_{lcs}^3 + \alpha_s d_{lcs}^4 - Y_{jics} \right) + \\
& \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C Cc_{lc} \cdot \left((1 - \alpha_s) d_{lcs}^3 + \alpha_s d_{lcs}^4 - Y_{ljcs} \right)
\end{aligned} \quad (33)$$

$$\begin{aligned}
E[Z_s] = & \sum_{j=1}^J \sum_{k=1}^K F_{jk} \cdot X_{jk} + \sum_{n=1}^N F_n \cdot U_n + \sum_{j=1}^J \sum_{c=1}^C h_{jc} \cdot J_{jc} + \sum_{j=1}^J \sum_{l=1}^L \sum_{c_1=1}^{C_1} b_{lc_1} \cdot Q_{ljc_1} + \\
& \sum_{j=1}^J \sum_{l=1}^L \sum_{c_2=1}^{C_2} b_{lc_2} \cdot Q_{ljc_2} + \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C \sum_{v_1=1}^{V_1} Tr_{ljcv_1} \cdot Q_{ljc} + \sum_{j=1}^J \sum_{l=1}^L \sum_{c_2=1}^{C_2} P_s \cdot b_{lc_2s} \cdot Q_{ljc_2s} + \\
& \sum_{j=1}^J \sum_{c=1}^C P_s \cdot h_{jc} \left(I_{jcs} - \sum_{i=1}^I Y_{jics} \right) + \sum_{s=1}^S P_s \sum_{j=1}^J \sum_{l=1}^L \sum_{c_1=1}^{C_1} b_{lc_1s} \cdot Q_{ljc_1s} + \\
& \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C P_s \left[\left(\frac{1-\lambda}{2} \right) (Tr_{ljcs}^1 + Tr_{ljcs}^2) + \left(\frac{\lambda}{2} \right) (Tr_{ljcs}^3 + Tr_{ljcs}^4) \right] \cdot Q_{ljcs} + \\
& \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C P_s \left[\left(\frac{1-\lambda}{2} \right) (Tr_{ljcs}^1 + Tr_{ljcs}^2) + \left(\frac{\lambda}{2} \right) (Tr_{ljcs}^3 + Tr_{ljcs}^4) \right] \cdot Y_{ljcs} + \\
& \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C P_s \left[\left(\frac{1-\lambda}{2} \right) (Tr_{ljcs}^1 + Tr_{ljcs}^2) + \left(\frac{\lambda}{2} \right) (Tr_{ljcs}^3 + Tr_{ljcs}^4) \right] \cdot Y_{jics} + \\
& \sum_{i=1}^I \sum_{h=1}^H P_s \left[\left(\frac{1-\lambda}{2} \right) (Tr_{ihs}^1 + Tr_{ihs}^2) + \left(\frac{\lambda}{2} \right) (Tr_{ihs}^3 + Tr_{ihs}^4) \right] \cdot XH_{ihs} + \\
& \sum_{i=1}^I \sum_{n=1}^N P_s \left[\left(\frac{1-\lambda}{2} \right) (Tr_{ins}^1 + Tr_{ins}^2) + \left(\frac{\lambda}{2} \right) (Tr_{ins}^3 + Tr_{ins}^4) \right] \cdot XN_{ins} + \\
& \sum_{n=1}^N \sum_{h=1}^H P_s \left[\left(\frac{1-\lambda}{2} \right) (Tr_{nhs}^1 + Tr_{nhs}^2) + \left(\frac{\lambda}{2} \right) (Tr_{nhs}^3 + Tr_{nhs}^4) \right] \cdot NH_{nhs} + \\
& \sum_{i=1}^I \sum_{g=1}^G P_s \left[\left(\frac{1-\lambda}{2} \right) (Tr_{igs}^1 + Tr_{igs}^2) + \left(\frac{\lambda}{2} \right) (Tr_{igs}^3 + Tr_{igs}^4) \right] \cdot XG_{igs} + \\
& \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C P_s \cdot Cc_{lc} \cdot \left((1 - \alpha_s) d_{lcs}^3 + \alpha_s d_{lcs}^4 - Y_{jics} \right) + \\
& \sum_{j=1}^J \sum_{l=1}^L \sum_{c=1}^C P_s \cdot Cc_{lc} \cdot \left((1 - \alpha_s) d_{lcs}^3 + \alpha_s d_{lcs}^4 - Y_{ljcs} \right), \forall s
\end{aligned} \quad (34)$$

Table 1 Numerical example size

Set	Size	Set	Size	Set	Size
I	5	C ₁	2	N	3
J	3	C ₂	2	H	3
K	4	S	2	G	4
L	4	C	4		

Analysis of the numerical example

In this part of the research, a numerical example has been solved in a small size corresponding to the sizes presented in Table 1.

This study presents a robust-fuzzy-probabilistic planning approach to optimize humanitarian relief logistics networks while considering the cost of deprivation. The proposed model accounts for the inherent complexities and uncertainties in disaster scenarios, including fluctuating demand, unpredictable supply chains, and the critical need to minimize human suffering. The methodology encompasses model formulation, data generation, and solution strategies, which are elaborated below.

Model formulation

The model integrates deprivation costs into a nonlinear optimization framework. The objective function is designed to minimize a weighted combination of deprivation costs, transportation costs, and facility operation costs. Constraints ensure equitable resource distribution, adherence to budgetary limitations, and timely delivery of aid. The model incorporates robust optimization techniques to address demand uncertainties, fuzzy logic to handle imprecise data, and probabilistic components to model scenario-dependent variability. This hybrid approach ensures flexibility and adaptability in real-world applications.

Table 2 Interval limits of deterministic parameters of the problem

Parameter	Interval limits	Parameter	Interval limits
θ_c	$\sim U(0.5 - 1)$	F_{jk}	$\sim U(10000 - 12000)$
δ_n	0.3	F_n	$\sim U(5000 - 9000)$
ω_j	0.4	b_{lc_1}	$\sim U(0.1 - 0.15)$
V_k	$\sim U(10 - 50)$	b_{lc_2}	$\sim U(0.1 - 0.15)$
$CapH_h$	$\sim U(1000 - 1200)$	h_{jc}	$\sim U(0.05 - 0.15)$
$CapG_g$	$\sim U(1000 - 1200)$	Tr_{ljc}	$\sim U(0.2 - 0.25)$
$CapN_n$	$\sim U(1000 - 1200)$	v_{lc_1}	$\sim U(40 - 45)$
σ_{lc_1}	$\sim U(0.1 - 0.15)$	v_{lc_2}	$\sim U(40 - 45)$
σ_{lc_2}	$\sim U(0.1 - 0.15)$	ωP_i	0.2
		Cc_{lc}	$\sim U(70 - 80)$

Table 3 Interval limits of uncertain parameters of the problem

Scenario 1					
Parameter	Interval limits		Parameter	Interval limits	
c_{jirs}	$\sim (0.07 - 0.10)$		v_{lc1s}	$\sim (30 - 35)$	
b_{lc1s}	$\sim (0.08 - 0.12)$		v_{lc2s}	$\sim (30 - 35)$	
b_{lc2s}	$\sim (0.08 - 0.12)$		w_{is}	$\sim (150 - 200)$	
			p_s	0.5	
	Level 1	Level 2		Level 3	Level 4
d_{ics}	$\sim (200 - 250)$	$\sim (250 - 300)$		$\sim (300 - 350)$	$\sim (350 - 400)$
Tr_{ljcs}	$\sim (0.2 - 0.25)$	$\sim (0.25 - 0.3)$		$\sim (0.3 - 0.35)$	$\sim (0.35 - 0.40)$
Tr_{lics}	$\sim (0.2 - 0.25)$	$\sim (0.25 - 0.3)$		$\sim (0.3 - 0.35)$	$\sim (0.35 - 0.40)$
Tr_{jics}	$\sim (0.2 - 0.25)$	$\sim (0.25 - 0.3)$		$\sim (0.3 - 0.35)$	$\sim (0.35 - 0.40)$
Tr_{ihis}	$\sim (0.18 - 0.22)$	$\sim (0.22 - 0.28)$		$\sim (0.28 - 0.32)$	$\sim (0.32 - 0.35)$
Tr_{ins}	$\sim (0.18 - 0.22)$	$\sim (0.22 - 0.28)$		$\sim (0.28 - 0.32)$	$\sim (0.32 - 0.35)$
Tr_{nhis}	$\sim (0.18 - 0.22)$	$\sim (0.22 - 0.28)$		$\sim (0.28 - 0.32)$	$\sim (0.32 - 0.35)$
Tr_{igs}	$\sim (0.18 - 0.22)$	$\sim (0.22 - 0.28)$		$\sim (0.28 - 0.32)$	$\sim (0.32 - 0.35)$
Scenario 2					
Parameter	Interval limits		Parameter	Interval limits	
c_{jirs}	$\sim (0.10 - 0.13)$		v_{lc1s}	$\sim (35 - 40)$	
b_{lc1s}	$\sim (0.12 - 0.16)$		v_{lc2s}	$\sim (35 - 40)$	
b_{lc2s}	$\sim (0.12 - 0.16)$		w_{is}	$\sim (200 - 250)$	
			p_s	0.5	
	Level 1	Level 2		Level 3	Level 4
d_{ics}	$\sim (250 - 300)$	$\sim (300 - 350)$		$\sim (350 - 400)$	$\sim (400 - 450)$
Tr_{ljcs}	$\sim (0.25 - 0.3)$	$\sim (0.3 - 0.35)$		$\sim (0.35 - 0.40)$	$\sim (0.40 - 0.45)$
Tr_{lics}	$\sim (0.25 - 0.3)$	$\sim (0.3 - 0.35)$		$\sim (0.35 - 0.40)$	$\sim (0.40 - 0.45)$
Tr_{jics}	$\sim (0.25 - 0.3)$	$\sim (0.3 - 0.35)$		$\sim (0.35 - 0.40)$	$\sim (0.40 - 0.45)$
Tr_{ihis}	$\sim (0.22 - 0.26)$	$\sim (0.26 - 0.30)$		$\sim (0.3 - 0.35)$	$\sim (0.35 - 0.40)$
Tr_{ins}	$\sim (0.22 - 0.26)$	$\sim (0.26 - 0.30)$		$\sim (0.3 - 0.35)$	$\sim (0.35 - 0.40)$
Tr_{nhis}	$\sim (0.22 - 0.26)$	$\sim (0.26 - 0.30)$		$\sim (0.3 - 0.35)$	$\sim (0.35 - 0.40)$
Tr_{igs}	$\sim (0.22 - 0.26)$	$\sim (0.26 - 0.30)$		$\sim (0.3 - 0.35)$	$\sim (0.35 - 0.40)$

Data generation

Due to the lack of access to real-world disaster relief data, this study employs synthetic data generated using a uniform distribution function. The uniform distribution is selected for its simplicity and capacity to simulate various plausible scenarios. Key parameters, including demand, supply availability, transportation costs, and facility capacities, are generated within predefined ranges to mimic realistic disaster conditions. These parameters are calibrated to ensure a balanced representation of low-, medium-, and high-demand scenarios.

Solution strategy

The optimization model's nonlinear nature necessitates using advanced solvers capable of handling global optimization problems. For this purpose, the BARON solver in GAMS 23.5.2 software is employed. BARON is well-suited

for solving nonlinear programming problems, as it combines branch-and-bound techniques with deterministic global optimization algorithms. The solver identifies optimal or near-optimal solutions by efficiently exploring the solution space, ensuring the results are feasible and high-quality.

The solution process begins by inputting the generated data into the model and running the BARON solver under various disaster scenarios. The results are analyzed to evaluate the trade-offs between cost minimization and the timeliness of aid delivery. Sensitivity analysis assesses the impact of key parameters, such as demand variability and transportation delays, on the model's performance. Additionally, robustness tests are performed to validate the model's effectiveness under worst-case scenarios.

Tables 2 and 3 show the range limits of deterministic and non-deterministic parameters of the developed mathematical model.

Table 4 The number of people affected by accidents and transferred to temporary shelters, hospitals, and emergency rooms

	Demand point	Temporary emergency	Scenario 1	Scenario 2
XN_{ins}	1	2	30	47
	2	1		41
	2	3	35	
	3	1	40	46
	4	2	35	
	4	3		41
	5	1	40	
	5	3		42
	Demand point	Hospital	Scenario 1	Scenario 2
XH_{ihts}	1	2	62	96
	2	1	70	83
	3	1		94
	3	2	80	
	4	3	73	84
	5	1	80	
NH_{ihts}	5	3		86
	Temporary emergency	Hospital	Scenario 1	Scenario 2
	1	1	24	26
	2	2	19	14
	3	2	10	
	3	3		24
	Demand point	Shelter	Scenario 1	Scenario 2
XG_{igs}	1	1		95
	1	4	61	
	2	1		82
	2	2	70	
	3	2	80	92
	4	1		83
	4	2	71	
	5	3	80	84

Table 5 Amount of relief items purchased in the pre-disaster phase

Supplier	Distribution center	Eco-friendly		Environmentally incompatible	
		Product 1	Product 2	Product 3	Product 4
1	1	4.912	4.912	4.912	4.912
1	3	4.932	4.932	4.932	4.932
2	2	4.881	4.881	4.881	4.881

Table 6 Costs of the entire logistics network under the scenario and different uncertainty rates

$\alpha_2 = \alpha_1$	Total costs	$P_1 = 1 - P_2$	Total costs
0.1	327,091.440	0.1	480,670.6
0.2	353,683.256	0.2	467,488.3
0.3	370,095.073	0.3	449,255.3
0.4	392,175.312	0.4	428,288.6
0.5	413,098.706	0.5	413,098.7
0.6	433,512.671	0.6	400,348.7
0.7	458,774.201	0.7	387,443.7
0.8	479,684.340	0.8	376,958.9
0.9	493,514.672	0.9	364,473.6

Table 7 Costs of the entire logistics network under different values of deprivation cost

C_c %	Z1
50 – %	408,682.5
30 – %	409,745.2
10 – %	411,279.6
0	413,098.7
10 + %	418,328.7
20 + %	425,675.3
30 + %	429,744.6

Table 8 Costs of the entire logistics network under different values of model robustness

ξ'	Total cost	η	Total cost
1	413,098.7	2	413,098.7
10	426,841.7	4	418,223.8
20	462,846.8	10	429,476.6
100	497,842.0	15	433,674.8

After designing the numerical example and determining the interval limits of the problem's deterministic and non-deterministic parameters, the numerical example analysis was carried out. Also, due to the use of the robust-fuzzy-probabilistic optimization method to control the non-deterministic parameters of the problem, the effective technical coefficients are considered as $\xi' = 1, \omega = 2, \eta = 2, \lambda = 0.4$. After solving the problem, the objective function value equal to 413,098.71 has been obtained, which is discussed in the following decision variables.

Table 4 shows the number of people injured and transferred to other levels of the supply chain network.

Table 5 also shows the number of relief items purchased from suppliers by the relief distribution center in the pre-disaster phase.

Based on this, supplier numbers 1 and 2 supplied products before the disaster. Also, three centers, Nos. 1, 2, and 3, were built as temporary emergency centers and provided services to people affected by accidents.

The following discusses the problem's sensitivity analysis under different parameters. Due to the problem's nonlinearity, the process of changing the values of the objective functions of the problem under the uncertainty rate has been investigated. The previous section solved the numerical example, assuming an uncertainty rate of 0.5. Table 6 shows the total costs of the logistics network under different uncertainty rates and scenarios.

Increase in the uncertainty rate, and due to the increase in the number of people affected by the accident and the demand from the accident site for relief items, the costs related to the supply and transportation of the items and the injured have increased. Therefore, it can be seen that with the increase of the uncertainty rate of as much as 90%, the total costs have increased by 19.46%. On the other hand, the results of Table 6 show that with the increase in the probability of the second scenario, the costs of providing, producing, and distributing relief products have increased due to the rise in demand in the network.

On the other hand, in the presented model, it was impossible to meet all the demands of the affected areas from relief items due to the disruption. Therefore, the model tries to send the most relief items to the affected areas by considering the balance of the problem. Thus, if the aid distribution centers do not meet the demand of the affected areas, the exclusion fee will be applied. In this section, by examining the effect of increasing and decreasing the cost of deprivation on the network, its impact on the objective functions of the problem has been measured. Therefore, the deprivation cost value is considered 10, 30, and 50% lower and higher than the base value applied in random data. Table 7 shows the total cost of the logistics network under the change in the deprivation cost values.

According to Table 7's results, the model aims to increase the number of goods transferred to fully satisfy the demand by increasing the deprivation cost. Therefore, the expenses related to deprivation have decreased, but on the other hand, the costs associated with the transfer of products have increased. As a result, the cost of the entire relief logistics network has increased.

In this research, a robust-fuzzy-probabilistic optimization method is used to control the non-deterministic parameters of the model. In this method, to prevent fuzzy numbers and the uncertainty rate to justify the problem and guarantee the maximum demand of damaged points, the robust method is used in different scenarios.

Therefore, the change in the most critical parameters related to the robustness of the model will lead to a change in the total costs and decision variables. Table 8 shows the relief logistics network's total costs in different optimization parameter values.

By examining the effects of the weighting coefficient of the objective function on the values of the objective function of the problem, it can be seen that with the increase of this weighting coefficient, the value of the objective function has increased due to the rise in the volume of transfers of relief items. Also, the results of Table 8 show that by increasing the penalty cost of demand estimation, the model tries to estimate the maximum demand of the affected points by increasing the costs of distribution, transmission, etc. This has led to increased transfer of relief items between different centers.

Pandemics pose significant challenges to humanitarian logistics, particularly in the timely distribution of medical supplies, protective equipment, and vaccines. In such scenarios, the proposed model can play a critical role in mitigating deprivation costs, which are exceptionally high due to the direct impact of delayed medical aid on mortality and morbidity rates. The model can simulate varying infection rates across regions, incorporating probabilistic elements to account for uncertainties like sudden demand spikes or supply chain disruptions caused by overwhelmed infrastructure or labor shortages.

Using a robust-fuzzy-probabilistic framework, the model prioritizes allocating limited resources to regions where the deprivation costs are most severe, such as areas with high population densities, low healthcare capacity, or delayed response times. By optimizing supply distribution routes and facility operations, the model ensures equitable access to critical resources while minimizing total costs. For instance, vaccine distribution during a pandemic can benefit from the model's ability to handle fluctuating demand, ensuring that vulnerable populations are prioritized.

This application also allows sensitivity analysis to assess the impact of various constraints, such as supply delays or resource shortages. This provides actionable insights for policymakers to improve resilience and effectiveness in managing pandemic logistics.

Conclusion

The importance of rescue and relief of the injured during an accident caused this research to model humanitarian aid logistics under uncertainty. The designed logistics network model minimizes the costs of the entire relief logistics network. The most important decisions taken in this model are determining the optimal location of temporary emergencies and distribution centers for the distribution of relief items, determining the optimal amount of accumulated relief goods in the distribution centers, and determining the

optimal amount of allocation of the transfer flow of goods and casualties between facilities, optimal routing. Since the model's parameters are defined as fuzzy numbers in different scenarios, the new robust-fuzzy-probabilistic method was used to control the parameters and transform the non-deterministic model into a solvable deterministic model. By analyzing the sensitivity of various parameters of the problem, it was also observed that with the increase in the uncertainty rate and as a result of the increase in the number of people affected by the accident and the demand for relief items at the accident site, the costs related to the supply and transportation of the items and the injured have increased. Also, by examining the effect of deprivation cost on the results of the model, it was observed that with the increase of deprivation cost, the model aims to increase the amount of goods transferred to satisfy the demand fully. Therefore, the expenses related to deprivation have decreased, but on the other hand, the costs associated with the transfer of products have increased. As a result, the cost of the entire relief logistics network has increased. Finally, by examining the effects of the weighting coefficient of the objective function on the values of the objective function of the problem, it can be seen that with the increase of this weighting coefficient, the value of the objective function has increased due to the increase in the volume of transfers of relief items.

While offering a robust framework for humanitarian relief logistics, this research has certain limitations. First, the reliance on synthetic data generated via uniform distribution limits the realism of the analysis, as actual disaster scenarios often exhibit complex, region-specific demand patterns and supply constraints. Second, the nonlinear nature of the optimization model, despite being handled by the BARON solver, can lead to increased computational complexity, particularly for large-scale problems with numerous decision variables and scenarios. Third, the focus on probabilistic and fuzzy methods introduces model assumptions that may oversimplify real-world uncertainties, such as dynamic changes in transportation networks or sudden shifts in demand. Additionally, the model does not fully account for behavioral factors like community responsiveness or organizational collaboration challenges. Lastly, while deprivation costs are central to the model, quantifying them accurately in practice can be challenging due to the lack of universally accepted metrics and data.

The researchers offer the following suggestions to improve the model and develop it:

- Using reliability in the correct transfer of relief items between centers
- Using the routing of helicopters to distribute relief items instead of vehicles
- Using precise and innovative methods in problem-solving

Authors' contributions

All authors contributed to the study's conception and design. JGN, HN, and ASJ performed material preparation, data collection, and analysis. All authors wrote the first draft of the manuscript, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability

This is not applicable. We do not analyze or generate datasets because our work proceeds using a theoretical and mathematical approach.

Declarations

Competing interests

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